

Chromosome-scale genome assembly of the European common cuttlefish *Sepia officinalis*

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eLife Assessment

This manuscript reports a high-quality genome assembly of the European cuttlefish, *Sepia officinalis*, a representative species of the Cephalopod lineage. The data are based on current best practices for sequencing and genome assembly, including PacBio HiFi long reads and Hi-C chromatin conformation capture; the analysis is currently in parts **incomplete**, as further analyses are required to confirm the correct chromosome number. This genome will be a **useful** resource for the community of researchers interested in cuttlefish biology and comparative genomics in general.

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Abstract

Coleoid cephalopods, a subclass of mollusks, exhibit remarkable adaptations, including the largest brains among invertebrates, camera-like eyes, and a distinctive embryonic development. They possess an advanced behavioral repertoire including dynamic camouflage.

The common cuttlefish *Sepia officinalis* has served as a key model organism in various research fields, spanning biophysics, neurobiology, behavior, evolution, ecology and biomechanics. More recently, it has become a model to investigate the neural mechanisms underlying cephalopod camouflage, using quantitative behavioral approaches alongside molecular techniques to characterize the identity, evolution and development of neuronal cell types.

Despite significant interest in this animal, a high-quality, annotated genome of its species is still lacking. To address this, we sequenced and assembled a chromosome-scale genome for *S. officinalis*. The final assembly spans 5.68 billion base pairs and comprises 47 repeat-rich chromosomes. Gene linkage analysis confirms the existence of 47 chromosomes, revealing clear homologies with related species such as *Euprymna scolopes* and *Doryteuthis pealeii*. Our work includes a comprehensive gene annotation and full-length transcript predictions that should be helpful for further evolutionary and single-cell expression studies. This genome

provides a valuable resource for future research on the evolution, brain organization, information processing, development, and behavior in this important clade.

Introduction

Coleoid cephalopods (octopus, squid, cuttlefish) are a highly derived group of mollusks, characterized by the largest nervous systems among all invertebrates (ca. 500 million neurons in an adult octopus^{1,2}, ca. 200 million neurons in adult cuttlefish) and specializations with a great historical importance for neuroscience (e.g., “giant axons”³ and “giant synapses”^{4–6}). These animals exhibit very sophisticated behaviors such as dynamic camouflage^{7–13}, learning^{14–16}, social communication^{17–19} and hunting^{20–22} as well as two-stage sleep^{23–25}. Because of these characteristics, cephalopods have been the focus of many fundamental studies by biologists, biophysicists and physiologists over the past century^{26–35}. Thanks to advances in sequencing technologies, coleoid cephalopods are now also emerging as animal models in molecular neurobiology, evolution and genomics³⁶. Recent studies in particular examined cephalopod biology from the perspectives of single-cell gene expression^{37–40}, genome topology and gene regulation⁴¹.

Despite the technical progress, the genomes of cephalopods remain challenging to assemble because of their large sizes and high repeat fractions, creating a bottleneck for further molecular studies. While the genomes of *Octopodiformes* (Octopus, Eledone, Argonauta) are often comparable in size to that of humans (around 3 Gigabases (Gb)^{42,43}) or possibly even smaller⁴⁴, the typical genomes of *Decapodiformes* (squids and cuttlefish) are often twice as large, reaching 6 Gb^{45,46}. The biggest contributing factors to this genome expansion are transposable elements (TEs), with different TE classes differentiating squid and cuttlefish genomes from those of Octopodiformes⁴⁷. Besides such differences in genome sizes, karyotypes in coleoid cephalopods are also unusual when compared to those of other invertebrates, including mollusks: for example, haploid chromosome numbers are around 30 in octopuses and around 46 in squids and cuttlefish^{46,48}, while they are between 8 and 19 in non-cephalopod mollusks^{49–56}. Karyotype reconstruction and validation techniques in animals with such large genomes are also not as developed as they are for more familiar invertebrate and vertebrate species.

The common cuttlefish *Sepia officinalis* has been used as a model organism for studies in neural control and development^{57,58}, behavior^{7,15}, evolution⁵⁹, environmental studies^{60,61} and biomechanics^{62,63}. Found in the Mediterranean, the English Channel and the European Atlantic, it can produce large egg clutches containing hundreds of embryos, several times throughout its semelparous reproductive cycle^{64,65}. More recently, this species has emerged as an important model to study the neural basis of cephalopod camouflage^{66,67} building on extensive behavioral studies^{7–13} and neuronal cell type characterization, evolution and development^{68–70}.

Despite a widespread interest in this animal, including the recent sequencing of several cephalopod species as part of the Aquatic Symbiosis Genomics project⁷¹, a high-quality, annotated genomic resource for this species is still lacking. We describe here the sequencing, assembly, and annotation of the *Sepia officinalis* genome. Our analysis provides evidence for the existence of 47 chromosomes, with clear homologies to the chromosomes of related species such as *Euprymna scolopes* and *Doryteuthis pealeii*. We predict full-length transcripts and provide functional gene annotation that should inform future evolutionary and transcriptomic studies. Taken together, this genome should be an important resource for scientists interested in genome evolution, brain organization, information processing, development and behavior.

Results

We sequenced the genome of a 6-month-old *Sepia officinalis* individual (probably male, see below), reared from eggs collected in the Portuguese Atlantic. The DNA was extracted from brain tissue and sequenced using long-read (PacBio HiFi) and chromatin conformation (Hi-C) methods (**Figure 1A,B**).

Genome size and heterozygosity

The haploid genome size of *S. officinalis* was estimated to be ~5.14 Gb based on k-mer estimation from the long-read data, with a high repeat content of 54 %^{72,73}. The heterozygosity rate was estimated to be 1.03 %, which is higher than in octopus genomes, yet moderate among marine invertebrates^{42,43,46}.

The size of the scaffolded assembly was 5.68 Gb (**Figure 1C**), thus about 10% greater than our initial GenomeScope estimate. Indeed, the sizes of most published metazoan genome assemblies deviate by more than 10% from estimates, with proportionally greater deviations as assembly sizes increase⁷⁴.

Completeness of the genome was assessed using BUSCO⁷⁵ with *metazoa odb10*. It was 95.2% [single copy: 93.5%, duplicated: 1.7%], with 3.2% fragmented and 1.6% missing BUSCOs. The BUSCO score with *mollusca odb10* for the final assembly was 77.2% completeness [single copy: 75.9%, duplicated: 1.3%] with 4.3% fragmented and 18.5% missing BUSCOs.

Karyotype

The assembly was generated from PacBio HiFi reads and Hi-C reads with ca. 23-fold and 11-fold coverage, respectively. Scaffolding was performed using Hi-C reads, placing 95.8 % of bases in 47 scaffolds. The Hi-C heatmap in **Figure 1D** shows the chromatin contacts in clusters corresponding to these 47 scaffolds. It is consistent with the proposal that they each represent a single chromosome.

To increase confidence in our assembled *Sepia officinalis* karyotype, we used several scaffolding programs and manually curated the scaffolds using two independent approaches. The first used HapHiC⁷⁶, a tool based on Hi-C data that utilizes allele information from primary assembly programs, and scaffolds assemblies with the constraint of an expected number of chromosomes. The resulting Hi-C contact maps with 46, 47, 48, 49 and 50 chromosomes (**Supplementary Figure 1**) show support for 47 chromosomes; an input of 46 scaffolds resulted in one clear chromosome merger (blue arrowhead); higher input numbers resulted in false chromosome splits (black arrowheads, 48, 49 and 50 scaffolds).

Second, we performed manual curation of the scaffolding result obtained with YAHS⁷⁷ using JBAT⁷⁸. These two approaches, conducted independently and both based on Hi-C signal, converged on 47 chromosomal scaffolds.

Karyotype comparisons with other Decapods

Previous karyotyping studies in decapod cephalopods based on cytogenetic techniques provide widely varying estimates of chromosome numbers, reflecting the difficulty of resolving large chromosome numbers *in situ*. For example, 1n=46 chromosomes has been reported for two species of cuttlefish (*Sepia esculenta* and *Sepia lycidas*) and three loliginid squids⁷⁹; 1n=34 chromosomes has been reported for *Sepia arabica*⁸⁰ and 1n=24 chromosomes in *Sepia pharaonis*⁸¹. In *Sepia*

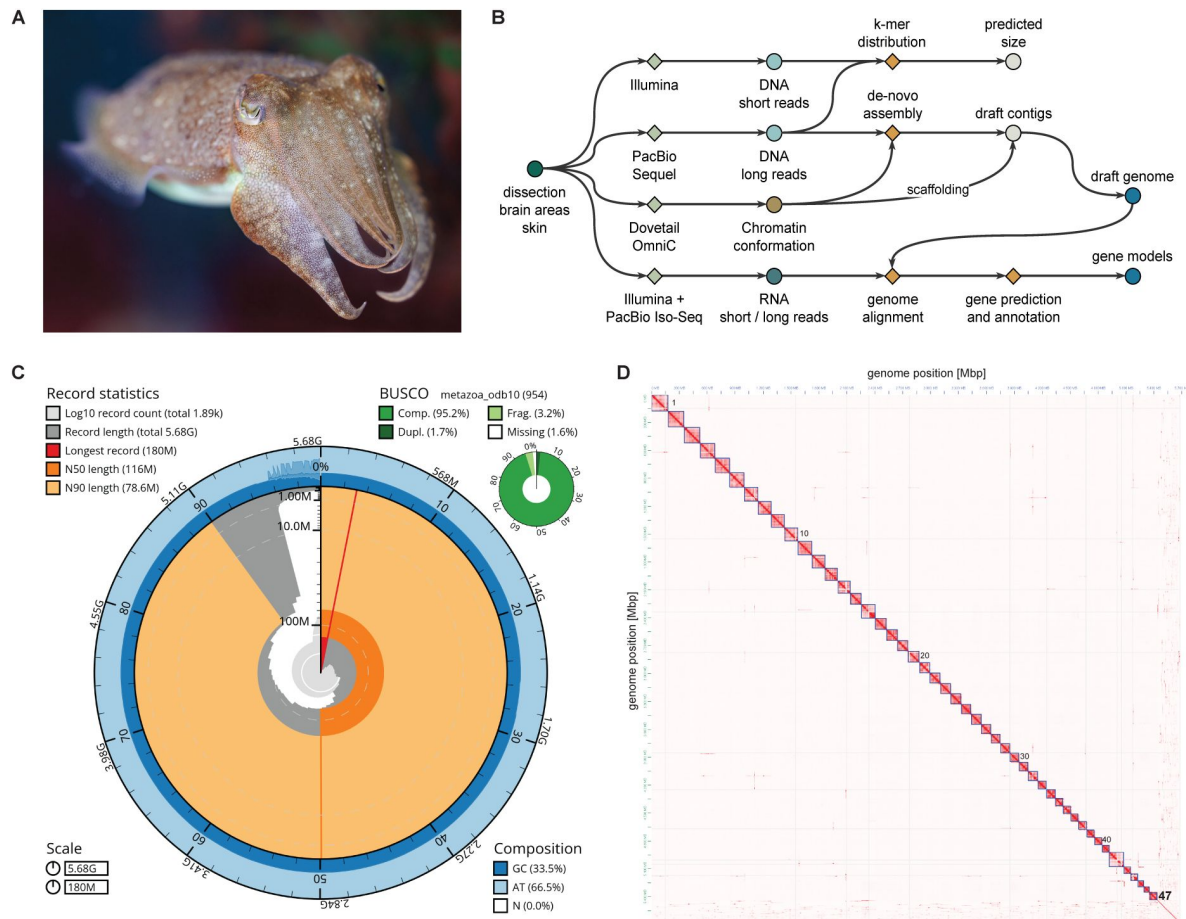


Figure 1

Sepia officinalis assembly statistics and quality control.

A) Specimen of *S. officinalis* (credit: Stephan Junek, MPI Brain Research). **B**) Overview of the assembly workflow. Genome size was estimated using GenomeScope^{72,73}. The primary assembly was generated from long DNA reads (PacBio Sequel II) and HiC reads (Dovetail OmniC) with hifiasm¹²⁴. Assembly was scaffolded with YAHS⁷⁷. **C**) Snail plot of chromosome-scale *S. officinalis* assembly generated using blobtools2¹³⁵ showing scaffold statistics, base composition and BUSCO⁷⁵ completeness against the metazoa_odb10 database. **D**) HiC heatmap showing 47 chromosome-scale scaffolds with very few sequences in unplaced scaffolds. Heatmap was generated using juicebox⁷⁸, 0-7039 observed counts (balanced) are shown.

officinalis, a karyotype of $1n=52$ has even been described from testis samples⁸². More recently, several cephalopod genomes have been scaffolded at chromosome-scale resolution, resolving karyotypes *in silico*. From those recent studies, a haploid karyotype of 46 chromosomes seems to be common among *Decapodiformes*⁴⁶ (**Figure 2A**).

Comparison with another chromosome-scale *Sepia officinalis* assembly

A chromosome-scale assembly for *Sepia officinalis* was released recently by the Wellcome Sanger Institute's Darwin Tree of Life project⁸³ (DToL, GCA_964300435.1). That genome was assembled from a male individual using PacBio Sequel II and Arima2 Hi-C data, with a final assembly size of 5.8 Gb. In that study, the haploid chromosome number was estimated to be 49. We scaffolded their assembly using our Dovetail Omni-C data, and the resulting contact map again supports 47 chromosomes (**Supplementary Figure 2A**), consistent with our results. We next aligned the two *S. officinalis* assemblies using minimap2⁸⁴ to assess the differences between them (**Supplementary Figure 2B**). The “Sanger” assembly appears to split three chromosomes of our assembly (chromosomes 1, 5 and 42) and to merge two (chromosomes 40 + 43). In conclusion, available evidence supports a haploid karyotype of 47 chromosomes in *Sepia officinalis*.

Comparisons with other chromosome-scale *Decapodiformes* genomes

To further investigate the possible reasons why our estimate of *Sepia officinalis*'s chromosome numbers deviates from those in other studies, we compared our assembly with the chromosome-scale genomes of two other decapod cephalopods, *Euprymna scolopes*⁴¹ (a *sepiolidae*) and *Doryteuthis pealeii*⁴⁶ (a *loliginidae*), both described as having 46 chromosomes. Using orthogroups to perform linkage analysis, we could detect a clear chromosome homology between *Doryteuthis* and *Sepia*, and to a lesser extent between *Euprymna* and *Sepia* (**Figure 2B**). Dotplots showing detailed pairwise synteny comparisons are shown in **Supplementary Figure 3** (*Sepia* to *Doryteuthis*) and **Supplementary Figure 4** (*Sepia* to *Euprymna*). The inferred species tree as part of this analysis places *Euprymna* as sister to *Sepia* and *Doryteuthis*, matching recent phylogenetic analyses^{50,85}. However, this tree was constructed without the inclusion of outgroup taxa, and therefore lacks a reliable root.

This comparison revealed that chromosomes 40 and 43 in *Sepia* are merged into one in *Euprymna* and in *Doryteuthis* (**Figure 2C**), as they were in the DToL *Sepia officinalis* assembly (**Supplementary Figure 2B**). Our scaffolding approach showed no contact between chromosomes 40 and 43 in our *Sepia* Hi-C data, providing support for these chromosomes being separate in *Sepia officinalis*.

We also compared our assembly to a chromosome-scale *Sepia esculenta* assembly ($1n=46$, GCA_964036315.1), and recently annotated by genome liftover from *Sepia pharaonis*⁸⁶. That assembly, constructed from a female individual, showed a low read coverage for a putative sex chromosome in a ZZ/Z0 sex determination system⁸⁷. Our syntenic comparison indicated a strong homology between the putative Z chromosome of *Sepia esculenta* and chromosome 46 of *Sepia officinalis* (**Supplementary Figure 5A**). Because we did not record the sex of the *Sepia officinalis* animal used for genome sequencing and assembly, we tried to infer the sex by replicating the analysis used to identify the putative Z chromosome in *Sepia esculenta*⁸⁷. For this, we aligned short-reads (Illumina) from the same *Sepia officinalis* individual to the assembly and examined the normalized read coverage for each chromosome (**Supplementary Figure 5B**). In contrast to the low coverage observed in the female *Sepia esculenta* assembly (**Supplementary Figure 5C**), we observed no decrease in read coverage for chromosome 46, suggesting that, if the alleged sex chromosome identity is correct, then our material came from a male animal.

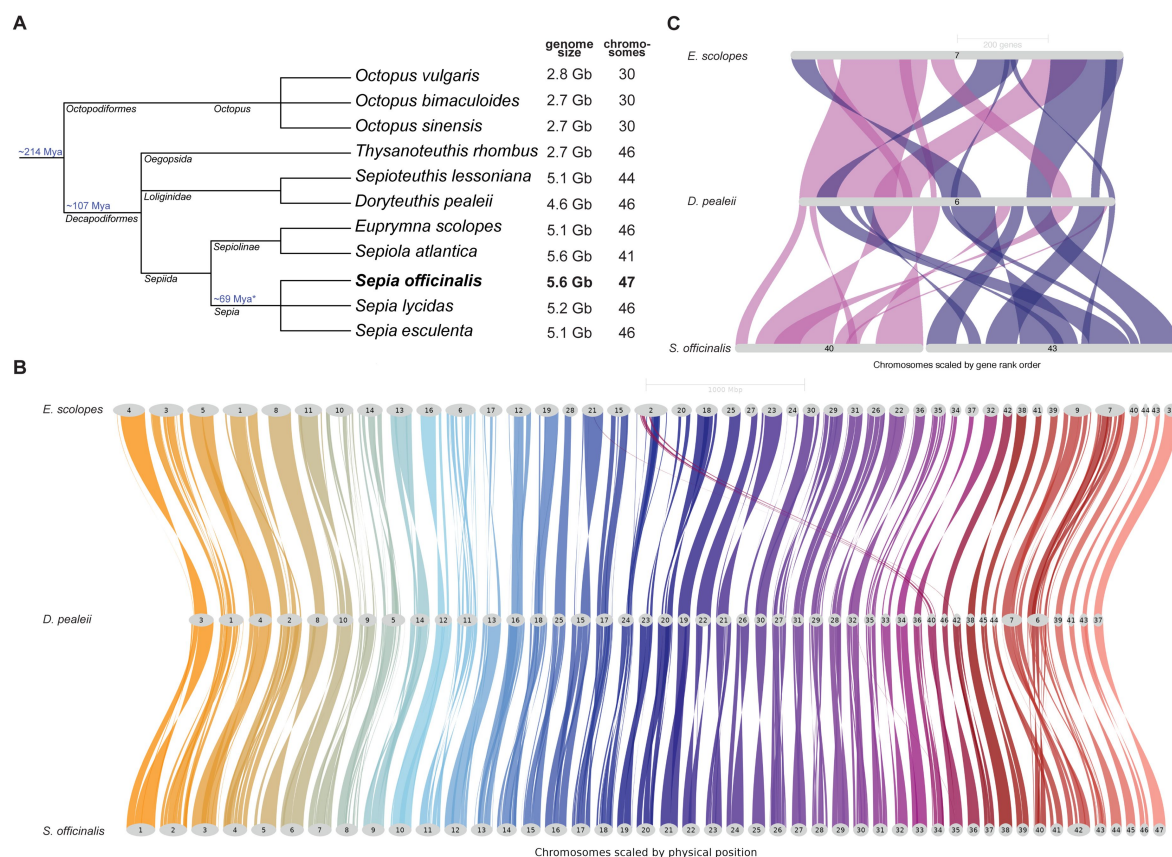


Figure 2

Syntenic comparison of three decapod species.

A) Taxonomy of selected cephalopod species showing their genome size and (haploid) chromosome numbers. Taxonomy information downloaded from NCBI taxonomy browser, divergence times from ⁸⁵[\[136\]](#). **B)** Synteny relationship between chromosomes of *E. scolopes*⁴⁶[\[46\]](#) (top), *D. pealeii*⁴⁶[\[46\]](#) (middle) and *S. officinalis* (bottom). Riparian plot was generated using orthogroups located on chromosomes of the species. *Euprymna* chromosomes 45 and 46 are not shown because they contain too few orthogroups. **C)** Synteny plot of *Sepia* chromosomes 40 (magenta) and 43 (darkblue), that are joined in the other species and cause the different haploid chromosome number in *Sepia*.

Genome repeat landscape

After creating a custom repeat library for *Sepia officinalis* using RepeatModeler⁸⁸, we masked the genome using RepeatMasker⁸⁹, resulting in 71.17% masked bases. The categories of repeats are shown in **Figure 3A**. Most repeats (39.65% of total bp) were not characterized and presumably represent ancient repeats that diverged beyond recognition⁹⁰. Retroelements constituted the largest characterized repeat category (17.32%) followed by DNA transposons (5.92%); 5.83% were annotated as simple repeats. As observed in other *Decapodiformes*, the *Sepia* genome contained almost no short interspersed nuclear elements (SINEs), supporting the hypothesis that the SINE expansion observed in octopuses⁴⁷ occurred independently in their lineage.

Gene modeling and annotation

The genome was further annotated using BRAKER^{75,91–106} combining short- and long-read RNA-seq data and publicly available protein data from multiple molluscan species (*Doryteuthis pealeii*⁴⁶, *Euprymna scolopes*¹⁰⁷, *Octopus bimaculoides*⁴⁶, *Octopus vulgaris*⁴³, *Nautilus pompilius*¹⁰⁸, and *Pecten maximus*⁵¹). A total of 18,663 gene models and 23,768 proteins were annotated. The gene annotation was evaluated using BUSCO⁷⁵ v5.5.0 in protein mode, showing high completeness of the annotation (metazoa_odb10: C:98.2%[S:77.3%,D:20.9%], F:1.0%, M:0.8%, n:954 and mollusca_odb10: C:81.5%[S:59.6%,D:21.9%], F:0.9%, M:17.6%, n:5295). We further checked the completeness and consistency of our gene models using OMArk¹⁰⁹, and compared them to genome annotations for two other cuttlefish species, *Sepia pharaonis* and *Sepia lycidas* (**Figure 3B**). The completeness is assessed by comparing the annotated genes to conserved orthologs (present in 80% of extant species) in a given taxonomic clade, here the superorder *Lophotrochozoa*. The *S. officinalis* annotation misses 181 out of 2373 genes, which is higher than *S. lycidas* (96 genes) but much lower than *S. pharaonis* (458 genes).

In a consistency assessment, where the presence of known gene families from the lineage is evaluated (**Figure 3C**), *S. officinalis* contains low proportions of taxonomically inconsistent or fragmented proteins (ca. 13%) similar to *S. lycidas* (9%), giving confidence in the annotation. In contrast, more than 50% of *S. pharaonis* proteins are labeled as inconsistent or fragmented, which could indicate annotation errors¹⁰⁹.

Overall, the annotations contain different numbers of predicted proteins, likely caused by differences in the annotation method and reference data used. *S. officinalis* contains the least proteins (23,768) compared to *S. lycidas* (35,949) and *S. pharaonis* (53,515). The *S. officinalis* annotation thus represents a conservative yet high-quality gene set. In the future, this annotation could be enhanced by incorporating more long-read RNA sequencing data, for example.

Lastly, we assigned orthology information to the *S. officinalis* proteome from InterPro¹¹⁰ and eggNOG-mapper¹¹¹ to aid the interpretability of the resource for future transcriptomic or proteomic studies. Overall, 89% of proteins (21,204 out of 23,768) received an annotation from InterPro from at least one of their databases. 59% of proteins (14,126 out of 23,768) were annotated by eggNOG-mapper, reflecting the more stringent orthology filters and prioritization of full-length matches implemented in the program¹¹². Similar to previous studies of other cephalopod genomes⁴², we observed a high number of C2H2 zinc finger transcription factors (1,538 proteins, InterPro annotation) and protocadherins (339 proteins, InterPro), suggesting a conserved expansion of these gene families among cephalopods. Further analyses should reveal more expanded gene families in *S. officinalis* that underlie species-specific adaptations.

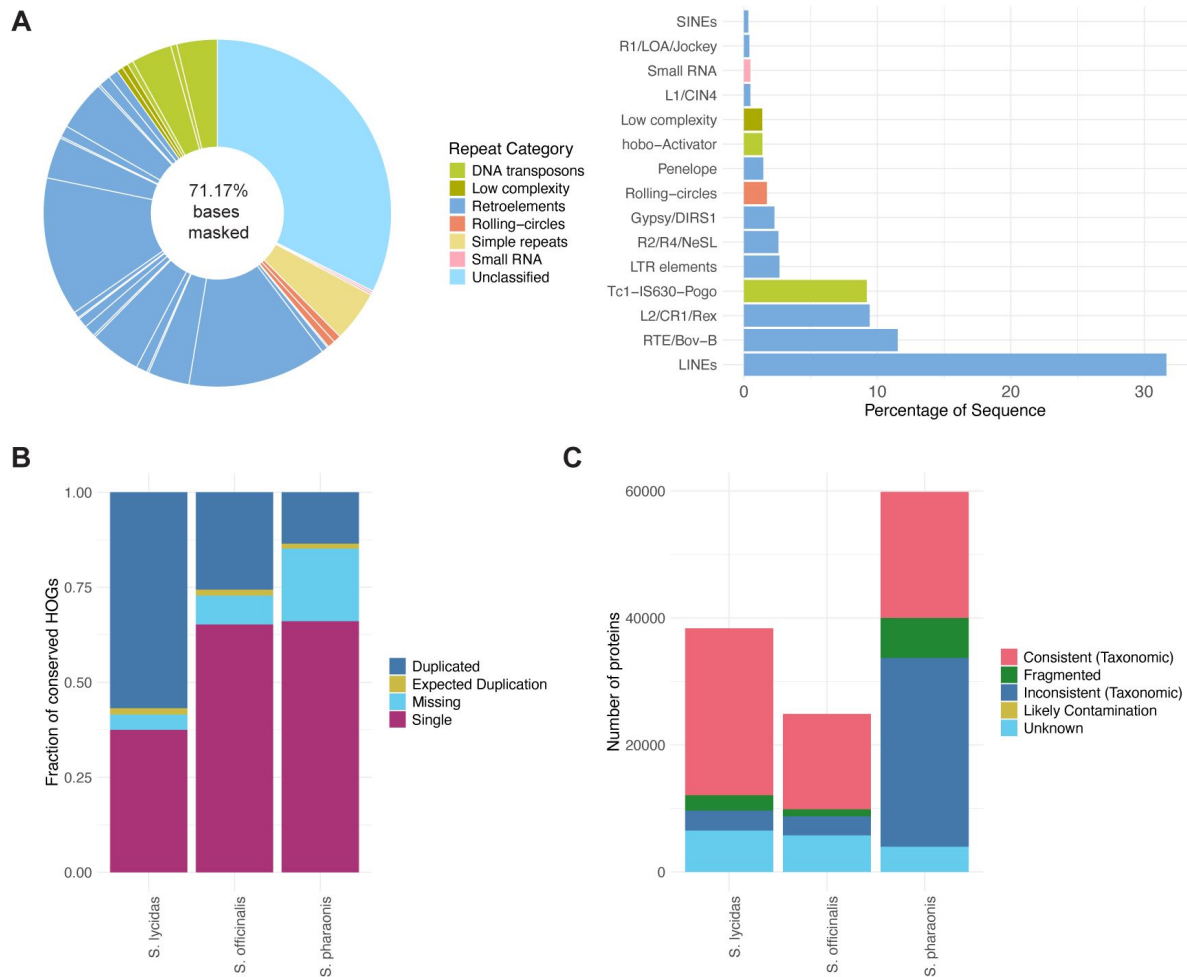


Figure 3

Genome annotation for *Sepia officinalis*.

A) Annotation of repeat landscape of the *S. officinalis* genome, annotated using RepeatModeler⁸⁸. Full repeat landscape is shown on the left, annotated repeats (excluding unclassified or simple repeats) are shown on the right. **B-C)** Quality control of gene annotation and comparison to two other cuttlefish species using OMArk¹⁰⁹. Results shown for *Sepia lycidas* (GCA_963932145.1, Ensembl Genebuild), *Sepia officinalis* (BRAKER, this study) and *Sepia pharaonis*¹³⁷ (BRAKER). *Lophotrochozoa* was used as the ancestral clade. **B)** Completeness assessed by the presence of genes conserved in the clade, classified as *single* or multiple copies (*duplicated*), or *missing*. **C)** Consistency assessed by the proportion of proteins placed in the correct lineage (*consistent*); placement in incorrect lineages randomly (*inconsistent*) or to specific species (*contamination*), or no placement in known gene families (*unknown*).

Discussion

This study presents a high-quality, chromosome-scale genome assembly and annotation for the common cuttlefish *Sepia officinalis*, providing an additional important resource for comparative genomics, neurobiology, and evolutionary studies within cephalopods and mollusks. The assembly was derived from a combination of PacBio HiFi long reads, Dovetail Omni-C chromatin conformation capture, and multiple rounds of scaffolding and manual curation, resulting in a genome size of 5.68 Gb and a haploid karyotype of 47 chromosomes. This resource represents a significant step forward for cephalopod genomics, especially within the *Decapodiformes*, where large and repetitive genomes have hindered previous efforts.

Chromosome Number and Karyotypic Variation

Our results strongly support a haploid chromosome number of 47 in *S. officinalis*, a conclusion reinforced by two independent scaffolding approaches (YAHSp⁷⁷ and HapHiC⁷⁶) and supported by Hi-C contact maps. This karyotype contrasts with the 46 chromosomes reported in other decapod cephalopods, such as *Euprymna scolopes* and *Doryteuthis pealeii*⁴⁶. Syntenic comparisons revealed that chromosomes 40 and 43 in *S. officinalis* correspond to a single chromosome in these other species. The current Hi-C data for *Sepia officinalis* provides clear evidence that chromosomes 40 and 43 are distinct, with no interaction contact. Because these two chromosomes are merged also in *S. esculenta*, it is plausible that this difference results from a *S. officinalis*-specific fission. However, our results show how difficult it still is to assemble large genomes with high karyotype numbers. Therefore, validation and comparisons with other *Sepia* species' genomes will be important. This, in turn, will help identify chromosomal fusion-with-mixing events that are clade specific. Early branching order in *Decapodiformes* has been notoriously unstable^{50,85,113,116}; thus, such rare and irreversible phylogenetic characters could be useful in further phylogenetic analysis of this clade^{48,117}.

We also compared our results to the recently published *S. officinalis* genome from the Darwin Tree of Life project (DTOL)⁸³, which proposed 49 chromosomes. The discrepancy is attributable to one merger in our assembly and three apparent chromosome splits in the DTOL assembly. While these differences could stem from technical choices such as different Hi-C methods (Arima v2 vs. Dovetail Omni-C) and sequencing depth (83-fold vs. 11-fold coverage), our assembly is strongly supported by Hi-C contact data, especially in regions where the DTOL assembly splits chromosomes that appear contiguous in our data. Interestingly, the DTOL assembly merges the chromosomes 40 and 43 - supporting a 46-chromosome model as ancestral state. However, this is not supported by re-scaffolding with the current Omni-C dataset, which again shows them as separate. This discrepancy might reflect technical differences, but it could also be hinting at population-level variability in chromosomal architecture; that is, some individuals may carry a fusion while others do not. Intraspecific chromosome number variation is rare but not unprecedented; for instance, chromosomal polymorphism has been described in the butterfly *Leptidea sinapis* across different populations^{118,119}. Notably, the two *S. officinalis* individuals used for sequencing originated from different regions - the Portuguese Atlantic (this study) and the French Mediterranean (DTOL project), raising the possibility of geographic variation in chromosome structure. Future population-level analyses will be critical to determine whether *S. officinalis* exhibits karyotypic polymorphism.

Genome Size and Repeat Landscape

The size of the genome of *S. officinalis* is estimated at 5.68 Gb - comparable to other *Decapodiformes* and roughly double the size of typical *Octopodiformes* genomes. Repeat annotation revealed that over 71% of the genome is repetitive, dominated by unclassified elements

and retrotransposons. The near absence of SINEs reinforces the idea that their expansion in octopuses was a lineage-specific event⁴⁷. These results underscore the complexity and evolutionary dynamism of the cephalopod genome, shaped by waves of transposon activity that likely played a role in the evolution of novel traits^{41,46}.

Gene Annotation and Comparative Assessment

We annotated 18,663 gene models and 23,768 proteins using a combination of short- and long-read RNA-seq data and molluscan protein references. Compared with other annotated cuttlefish genomes (*S. lycidas*, *S. pharaonis*), our annotation is conservative in gene number but stands out in terms of completeness and taxonomic consistency. OMark-based evaluations confirmed that our gene models have low levels of fragmentation and contamination, and high lineage-specific coherence. Striking differences across species in the number of predicted genes, duplications, and missing orthologs likely reflect variation in the annotation pipelines and reference data rather than true biological divergence. Still, our dataset provides a high-confidence foundation that can be refined with additional transcriptomic data from long-read techniques and diverse tissues.

As in other coleoid genomes, we observed notable expansions of gene families such as C2H2 zinc finger transcription factors and protocadherins, genes implicated in neural development and plasticity^{42,45}. These conserved expansions across cephalopods point to genomic underpinnings of their cognitive complexity and behavioral sophistication.

Conclusions and Outlook

This study delivers a foundational genomic resource for *Sepia officinalis*, a model for studying neural control, behavior, and adaptive camouflage. The chromosome-scale assembly and annotation open the door to in-depth studies of gene regulation, neural circuit development, and genome evolution across coleoid cephalopods. Moreover, the identification of possible population-level karyotype variation highlights the importance of incorporating broader sampling strategies in future genomic studies.

As we move toward a more complete picture of cephalopod genome evolution, the integration of chromosomal synteny, regulatory architecture, and transcriptomic diversity across species will be especially important. The *S. officinalis* genome represents an important step on this path, enabling high-resolution comparative and functional analyses of one of the most enigmatic and evolutionarily successful invertebrate lineages.

Materials & Methods

Animal husbandry

All research and animal care procedures were carried out following the institutional guidelines that comply with national and international laws and policies (DIRECTIVE 2010/63/EU; German animal welfare act; FELASA guidelines). European cuttlefish *Sepia officinalis* were hatched from eggs collected in the Portuguese Atlantic and reared in a seawater system, at 20°C. The closed system contains 4,000l of artificial seawater (ASW; Instant Ocean) with a salinity of 33‰ and pH of 8–8.5. Water quality was tested daily and adjusted as required. Trace elements and amino acids were supplied weekly. Marine LED lights above each tank provided a 12/12-h light/dark cycle with gradual on- and off-sets at 07:00 and 19:00. The animals were fed live food (either *Hemimysis* spp. or small *Palaemonetes* spp.) *ad libitum* twice per day. The animals were housed together in 120-l glass tanks with a constant water through-flow resulting in five complete water exchanges per hour. Enrichment consisted of natural fine-grained sand substrate, seaweed (*Caulerpa prolifera*), rocks of different sizes and various natural and man-made three-dimensional objects.

Tissue preparation

Animals underwent terminal anesthesia to prevent animal suffering or distress, following the Guidelines for the Care and Welfare of Cephalopods in Research^{120,121}. Animals were transferred to a bucket containing ethanol 2% (v/v) in ASW. After 5 minutes, animals were gently probed for simple avoidance reflexes, and the ethanol concentration in the ASW was gradually increased to a maximum of 5%. Sufficient depth of anesthesia was reached when the animal no longer reacted even to stronger stimuli (touching and pinching with tweezers), and no reaction was observed when a hand was moved in the visual area and the cornea was touched¹²². In this deep anesthesia, the animals were decapitated and the tissue was rapidly transferred into ice-cold, calcium-free Ringer solution (460 mM NaCl, 10 mM KCl, 51 mM MgCl₂, 10 mM glucose, 2 mM Glutamine, 10 mM HEPES, pH 7.4) bubbled with oxygen. Brain and body tissue was dissected under a stereoscope and flash-frozen in liquid nitrogen before storage at -80°C until further use.

Unless stated otherwise, sequencing libraries were prepared from the same individual (6-month-old adult *Sepia officinalis*, F1 from eggs collected in Portugal).

Long-read Whole Genome Library Preparation and Sequencing

The long-read library was prepared and sequenced at the MPI for Plant Research in Cologne, Germany. Genomic DNA was extracted from flash-frozen brain tissue with the MagAttract HMW DNA kit (Qiagen, Cat. no. 67563) and the sequencing library was prepared with the SMRTbell express template prep kit 2.0 (PacBio, Cat. no. 100-938-900). The library was sequenced on 5 SMRT cells on the PacBio Sequel II with the Sequel II binding kit 2.2 (102-089-000), Sequel II sequencing kit 2.0 (101-820-200) and SMRT Cell 8M tray (101-389-001).

Long-read RNA library preparation and sequencing

RNA was isolated from 5-10 mg of flash-frozen tissue (brain, mantle/epidermis, arm/tentacle). First, tissue samples were homogenized in TRIzol using a homogenizer (VDI12/S12N-5S, VWR, Germany). RNA was isolated and DNaseI treated according to the Direct-zol RNA Mini prep kit from ZymoResearch (R2050). RNA was quantified by spectrophotometry (Nanodrop, Thermo Scientific, USA) and its quality was assessed with a RNA 6000 Nanochip on Bioanalyzer (Agilent Technologies, Germany).

The Iso-Seq libraries were prepared and sequenced at the Sequencing facility of the MPI for Plant Research, using a method targeting the 5' cap and poly-A tail of mRNAs, described in detail in¹²³. Briefly, mRNA was pooled from the 8 tissue samples and cDNA was synthesized using the TeloPrime Full-Length cDNA Amplification Kit V2 (Lexogen, Cat Nr: 013.08 or 013.24). The sequencing library was prepared with the SMRTbell express template prep kit 2.0 (PacBio, Cat. no. 100-938-900). The pooled library was sequenced on 1 SMRT cell on the PacBio Sequel II system using the Sequel II binding kit 2.1 (101-843-000), Sequel II sequencing kit 2.0 (101-820-200) and SMRT Cell 8M tray (101-389-001).

Further, a separate library was prepared from optic lobe cDNA and sequenced on a second SMRT cell using the same reagents described above.

Omni-C Library Preparation and Sequencing

An Omni-C library was prepared and sequenced at the MPI for Plant Research in Cologne, Germany. The library was prepared from brain tissue using the Dovetail™ Omni-C™ Kit (Dovetail Genomics, Cat. No. 21005) according to the manufacturer's protocol. Briefly, chromatin was fixed in place in the nucleus. Fixed chromatin was digested with DNase I then extracted. Chromatin ends were repaired and ligated to a biotinylated bridge adapter followed by proximity ligation of adapter-containing ends. After proximity ligation, crosslinks were reversed and the DNA was

purified from proteins. Purified DNA was treated to remove biotin that was not internal to ligated fragments. The sequencing library was generated using Illumina-compatible adapters. Biotin-containing fragments were isolated using streptavidin beads before PCR enrichment of the library. The library was sequenced on an Illumina NextSeq 2000 platform to generate 400 million 2 x 150 bp read pairs.

Short-read DNA sequencing

High molecular weight genomic DNA was isolated from brain tissue (optic lobe) using NEB Monarch gDNA Purification Kit (T3010S). First, 10 mg of flash-frozen tissue were submerged in 100 µl of tissue lysis buffer, cut into small pieces and transferred into a 1.5 ml tube containing an additional 100 µl of lysis buffer. Then, 3 µl of 20 U/µl Proteinase K was added to the tissue sample and incubated for 2 h at 56°C in a thermal mixer with agitation until tissue pieces were completely dissolved. After incubation, tissue lysate was purified from the debris by centrifugation for 3 min at maximal speed (Eppendorf, Germany). Clean supernatant was transferred to a new 1.5 ml tube and incubated with 3 µl RNase A for 5 min at 56°C with agitation at full speed. After RNase A incubation tissue lysate was column purified and eluted with 80 µl of elution buffer according to the manufacturer protocol. The purity of gDNA was assessed using A260/280 and A260/230 absorbance ratios measured by NanoDrop spectrophotometer (ThermoFisher Scientific). The integrity of the obtained gDNA was checked using 0.75% agarose gel electrophoresis by loading 100 ng of gDNA sample and using Hind III digested Lambda DNA (NEB#3012) as a marker.

DNA sequencing libraries were prepared from 500 ng of high MW gDNA using Illumina DNA PCR-Free Tagmentation Library Prep Kit (20041794) and UD Indexes Set A (20026121). Obtained dual-indexed single-stranded libraries were quantified using the Qubit single-stranded DNA (ssDNA) assay kit (Q10212, ThermoFisher Scientific). 1000 pM of the final library were run on P3 reagents and Illumina NextSeq2000 using Illumina DNA PCR-Free Sequencing and Indexing primer (20041797) with 151 bp paired-end reads.

Short-read RNA library preparation and sequencing

For short-read RNA sequencing, tissue from another animal (8-month-old adult, F0 from eggs collected in Normandie, France) was used. RNA was isolated from 5 mg of flash-frozen tissues (brain, skin and retina). First, tissue samples were homogenized in TRIzol using a homogenizer (VDI12/S12N-SS, VWR, Germany) following DNaseI treatment and column purification according to the Direct-zol RNA Micro prep kit from ZymoResearch (R2062). The RNA integrity and quantity were measured with the Qubit fluorometer (Invitrogen, Q33216) and the 2100 Bioanalyzer (Agilent Technologies, Germany).

RNA-seq libraries were prepared from 300 ng of total RNA, using the Illumina TruSeq stranded mRNA library prep kit (Cat:20020594) and IDT for Illumina xGen UDI-UMI Adapters (Cat:10005903). Libraries were sequenced on an Illumina NextSeq500 (Mid output, 300 cyc) and NextSeq2000 (P3, 300 cyc) with 145 bp paired-end reads.

Nuclear genome assembly

Before assembly, PacBio HiFi reads were subjected to additional trimming to remove residual adapter sequences using NCBI's VecScreen tool. Genome size was estimated using GenomeScope⁷² from Illumina short reads and PacBio HiFi data. The primary assembly was generated with hifiasm¹²⁴ using a combination of HiFi and Hi-C reads.

Scaffolding was performed iteratively with YAHS⁷⁷ on the phased haplotype 1. We adjusted the run parameters for scaffolding resolutions (-r 1000,2000,5000,10000,20000,30000,50000,70000,100000,150000,200000,250000,300000,

350000,400000,500000,700000,1000000,2000000,5000000,10000000,15000000,20000000,30000000,40000000,50000000,60000000,70000000,80000000,100000000,110000000,120000000,150000000,170000000,200000000,500000000), repetitions per resolution (-R3), minimum read mapping quality (-q1) and telomeric sequence (--telo-motif "TTAGGG") from the only experimentally determined telomere motif for cephalopods (*Amphioctopus areolatus*¹²⁵, accessed via TeloBase¹²⁶) to optimize the tool's performance for a large, fragmented assembly. Finally, the assembly was manually curated using JBAT⁷⁸ to place residual scaffolds into chromosome scaffolds. Different versions of the assembly were evaluated based on HiC coverage, mRNA alignment and completeness based on BUSCO⁷⁵ v5.5.0 (*metazoa_odb10* and *mollusca_odb10*).

Mitochondrial genome assembly

To assemble the mitochondrial genome, we aligned a published *S. officinalis* mitochondrial genome¹²⁷ (NC_007895.1) to the PacBio Hi-Fi and Omni-C reads using minimap2⁸⁴. The hits were extracted using seqtk¹²⁸ subseq and assembled using hifiasm¹²⁴. The resulting contigs contained multiple repeats of the circular mitochondrial genome, and were aligned back to NC_007895.1 to extract the final sequence.

Nuclear genome annotation

Repetitive elements in the genome were softmasked using RepeatMasker v4.1.7-p1⁸⁹ (with rmbblast v2.14.1+) after creating a custom repeat library with RepeatModeler v2.0.6⁸⁸.

Gene models were created using BRAKER3^{75,91-106}, run via the Docker container, on the softmasked genome using both RNA-seq and protein data. We used long-read Iso-Seq data and short-read Illumina RNA-seq data generated for this study (see above). For protein data, publicly available peptide annotations for *Doryteuthis pealeii*⁴⁶, *Euprymna scolopes*¹⁰⁷, *Octopus bimaculoides*⁴⁶, *Octopus vulgaris*⁴³, *Nautilus pompilius*¹⁰⁸, *Pecten maximus*⁵¹ were used for training, as well as a previously generated proteome for *Sepia officinalis* from StringTie gene models (this study). RNA-seq data was input into BRAKER3 using the --bam option, protein files were specified with the --prot_seq option. The BUSCO completeness of the resulting gene set was maximized using TSEBRA within BRAKER on the BUSCO lineage *metazoa_odb10*.

For the initial protein set used as input for BRAKER, the short and long RNA reads were aligned to the genome using minimap2⁸⁴. Transcript models were predicted with StringTie v3.0.0¹²⁹ using the --conservative and --mix options. The resulting GTF files were combined using the transcript merge mode, resulting in a set of non-redundant transcripts. Finally, TransDecoder¹³⁰ v5.7.0 was run with default parameters to translate coding regions in the transcripts.

The final gene annotations for *S. officinalis* were assessed for completeness using OMArk v0.3.0¹⁰⁹ with the ancestral clade *Lophotrochozoa* and without including splice information, accessed via their webserver. Proteins were annotated with orthology information using InterProScan¹¹⁰ v5.73-104 including lookup of annotations and GO terms (options -iprlookup -goterms). Further orthology information was added using the eggNOG-mapper v2.1.12¹¹¹ webserver with the eggNOG v5.0 database and default parameters.

Whole genome alignment and syteny analysis

For whole genome alignments, the assembly produced for *Sepia officinalis* in this study and the published assembly from the Darwin Tree of Life project (GCA_964300435.1) were aligned using minimap2⁸⁴ and visualized with a custom script in R v4.4.2¹³¹.

For synteny analyses, peptide and gtf files were downloaded for *Doryteuthis pealeii* and *Euprymna scolopes*. Annotation files for *Sepia esculenta* were recently generated by liftover annotation from *Sepia pharaonis*⁸⁷ and downloaded from Zenodo⁸⁶. Synteny analyses between all chromosomes of the compared species were performed using the R package GENESPACE v.1.2.3¹³². *S. officinalis* was used as the reference species for all synteny maps.

Coverage analysis

Read coverage across chromosomes was analyzed as described recently⁸⁷. Short reads were aligned using STAR v2.7.11b¹³³ to our chromosome-scale assembly. For *Sepia esculenta*, short reads (ERR12945500) and assembly (GCA_964036315.1) were downloaded from NCBI. The sequencing depth was calculated using mosdepth¹³⁴ by a window size of 500,000 bp and normalized to the median coverage of the first chromosome.

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Additional information

Data availability

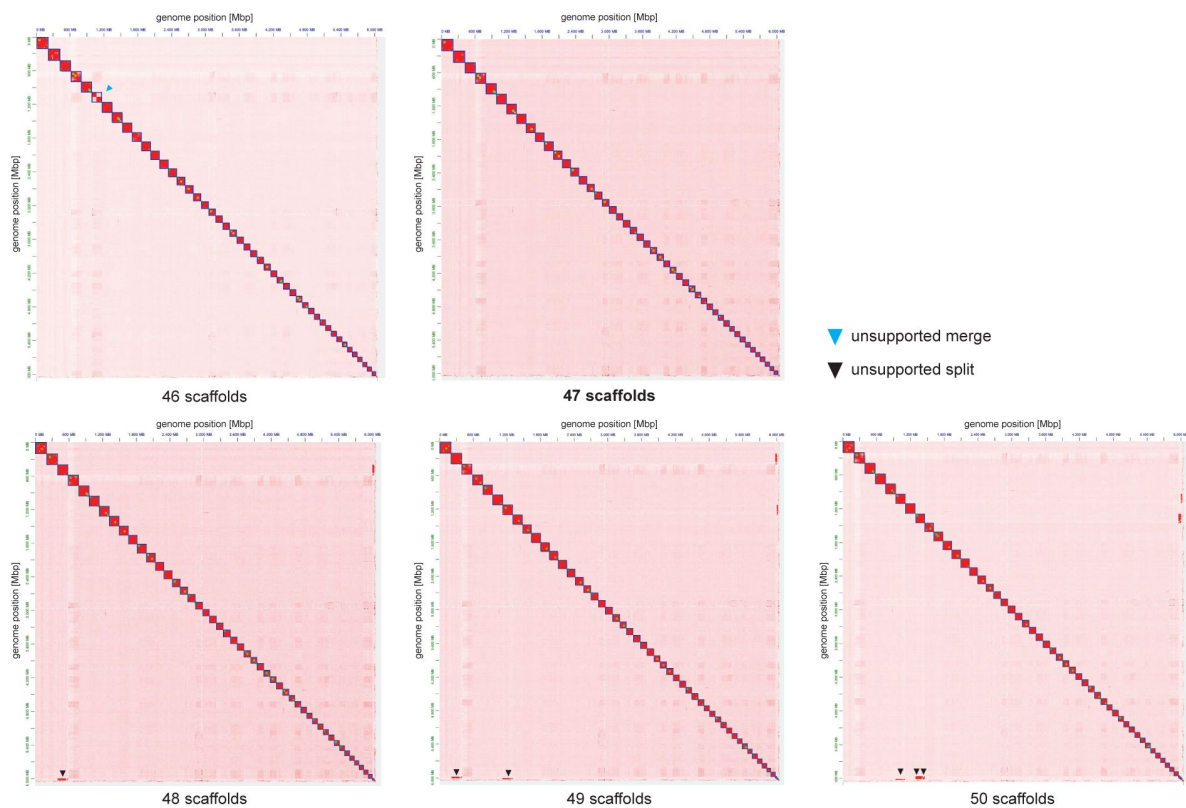
The genome assembly and raw sequencing reads can be found at the BioProject PRJNA1091451 on NCBI. The code for the genome assembly and annotation is available at <https://gitlab.mpg.de/mpibr/laur/cuttlefishomics/soffgenome>.

Genome annotation files are deposited at <https://public.brain.mpg.de/Laurent/sepoff/annotation/>.

Author contributions

O.S., and G.L. conceived of the study design. S.R., O.S. and G.L. wrote the first draft of the manuscript. D.H. collected tissue for sequencing. E.C. extracted DNA and RNA from tissue and performed Illumina sequencing. G.T. performed genome assembly. G.T. and S.R. manually curated and annotated the genome. S.R., G.T., and O.S. performed genomic analyses. S.R. created the figures. S.R., O.S. and G.L. coordinated and led the project. G.L. acquired funding for this study. All authors contributed to the review and editing of the manuscript.

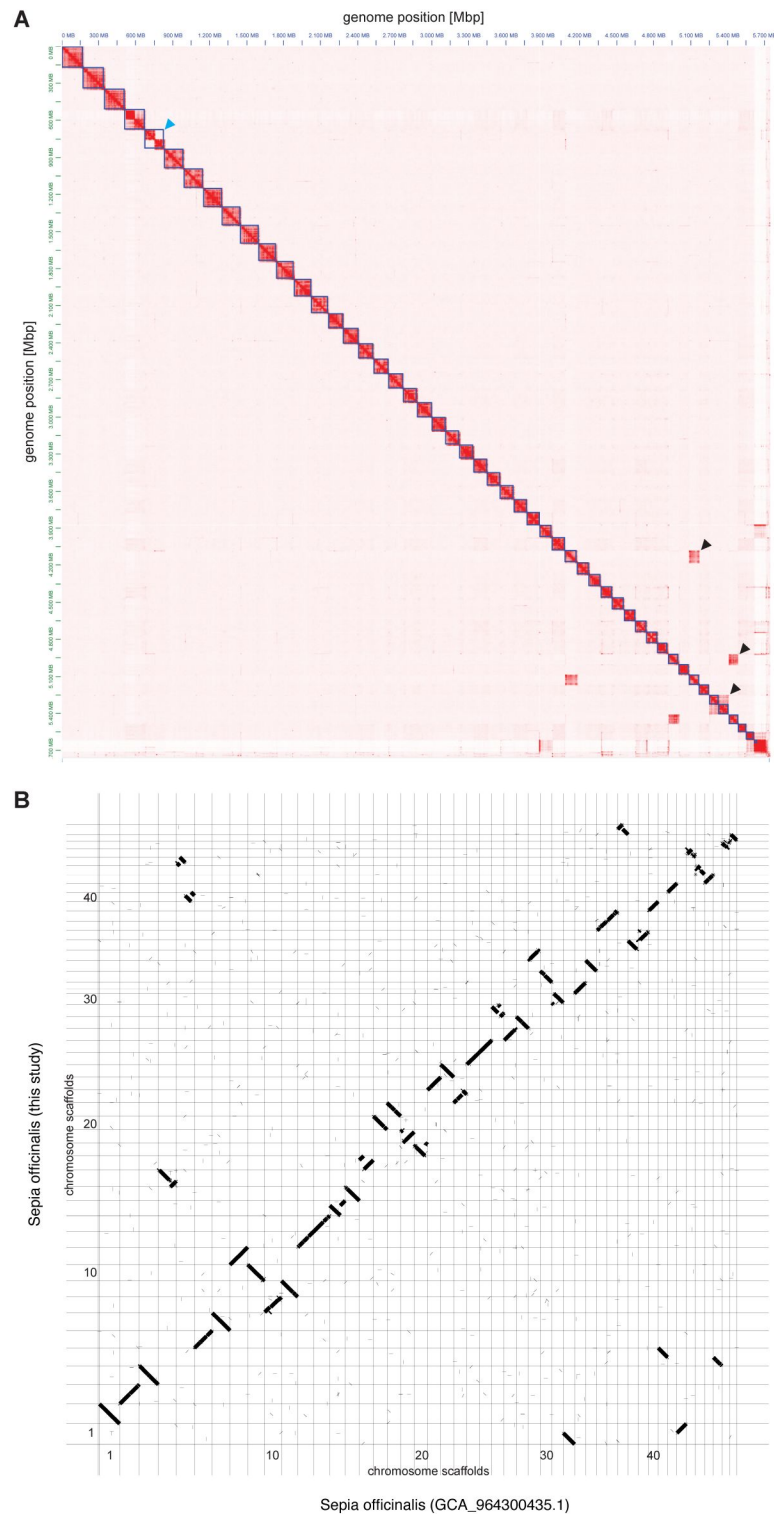
Funding



Supplementary Figure 1

HapHiC scaffolding for different numbers of expected chromosomes show 47 chromosomes as most supported.

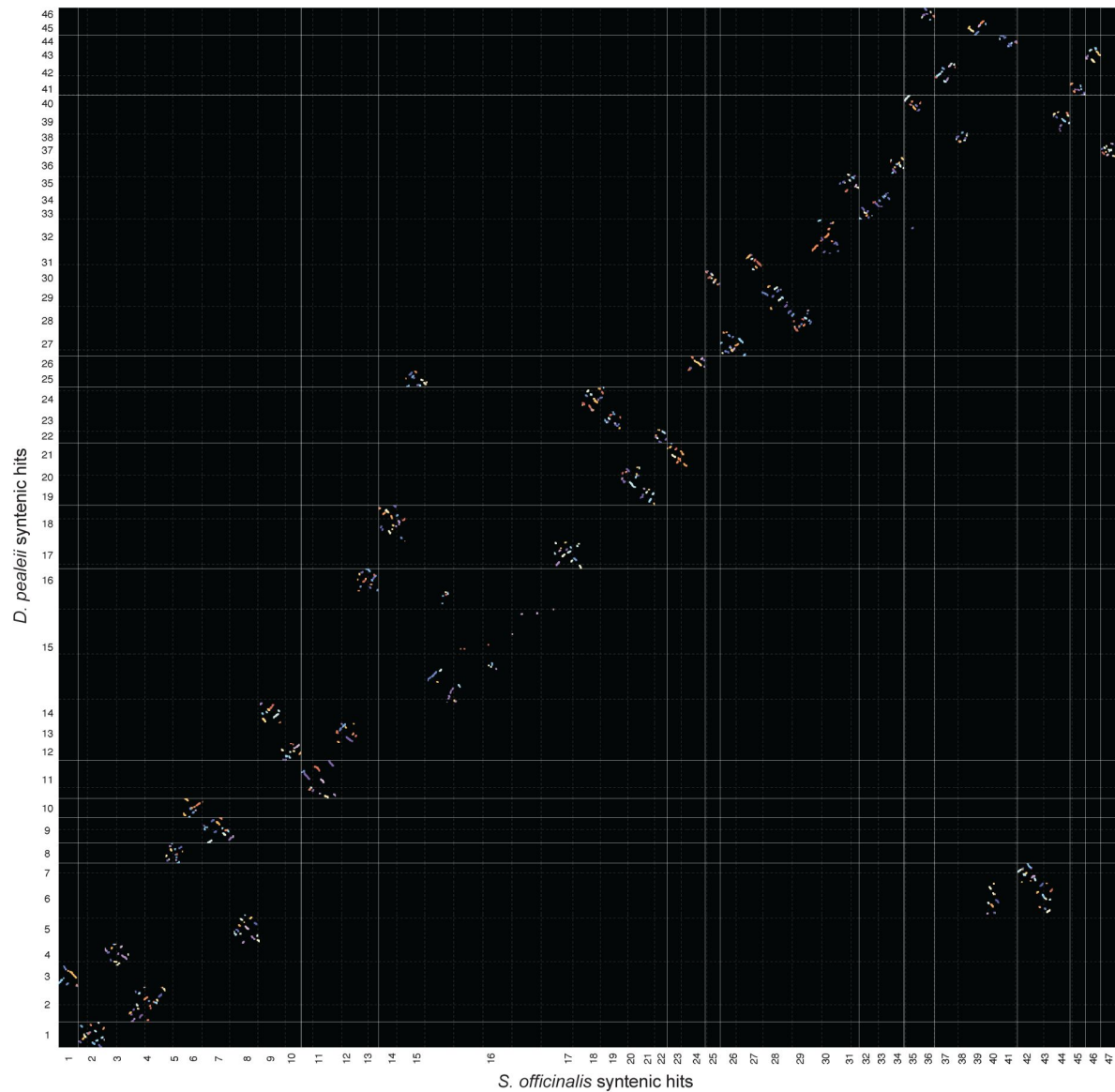
Hi-C contact maps from HapHiC⁷⁶ are shown for 46, 47, 48, 49 and 50 expected chromosome scaffolds. Assembled chromosomes are shown as blue boxes, HiC signal indicating a false (unsupported) merger is shown by cyan arrow, false splits are shown by black arrows.



Supplementary Figure 2

Comparison of *Sepia officinalis* chromosome-scale assemblies.

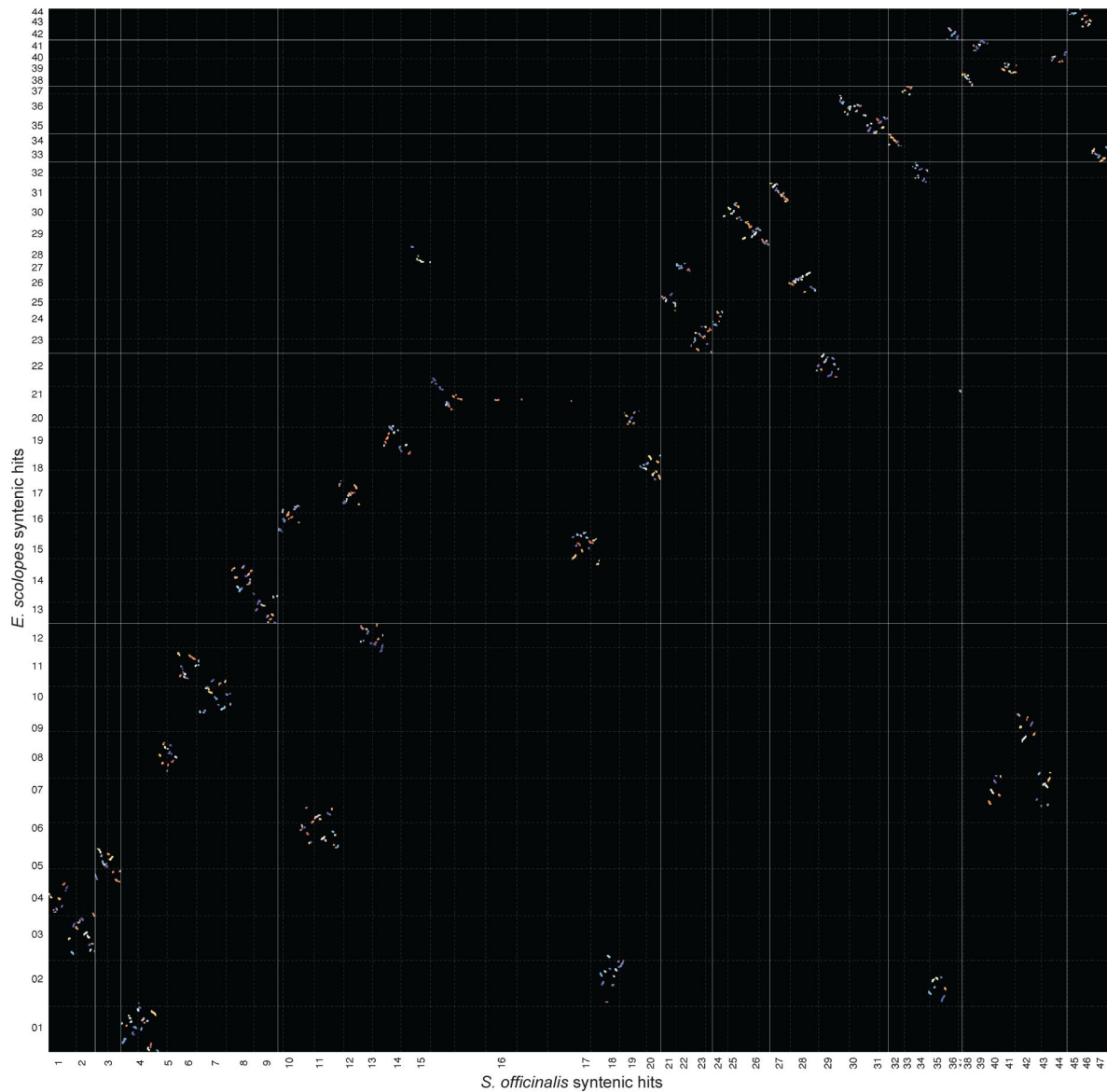
A) HiC heatmap of GCA_964300435.1 scaffolded against Omni-C data collected from a different individual (this study). Assembled chromosomes are shown as blue boxes, HiC signal indicating a false merge is shown by cyan arrow, false splits are shown by black arrows. **B)** Whole-genome alignment of both *S. officinalis* assemblies (this study: rows, GCA_964300435.1: columns).



Supplementary Figure 3

Syntenic relationship between *S. officinalis* and *D. pealeii* chromosomes.

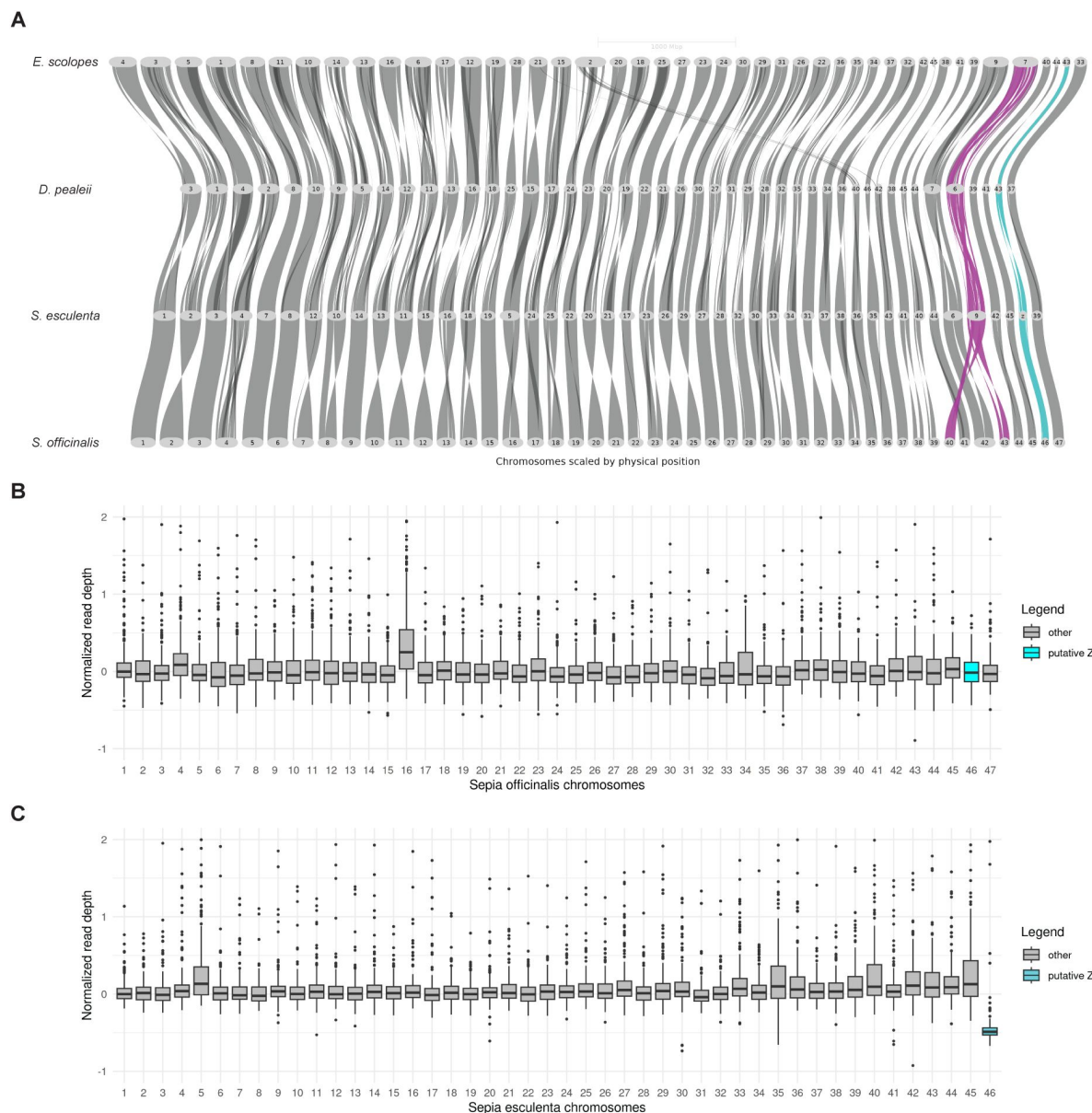
Dot plot showing finer-resolution syntenic anchor hits (perfectly collinear blast hits within the same orthogroup). Genes are ordered along the chromosomes, gridlines are shown every 1000 genes.



Supplementary Figure 4

Syntenic relationship between *S. officinalis* and *E. scolopes* chromosomes.

Dot plot showing finer-resolution syntenic anchor hits (perfectly collinear blast hits within the same orthogroup). Genes are ordered along the chromosomes, gridlines are shown every 1000 genes. *E. scolopes* chromosomes 45 and 46 are not shown because they contain too few orthogroups.



Supplementary Figure 5

Syntenic comparison of four decapod species hints at a cephalopod sex chromosome.

A) Riparian plot showing synteny relationships of chromosomes from four decapod species, generated using orthogroups. *Euprymna* chromosomes 45 and 46 are not shown because they contain too few orthogroups. Chromosome split in *S. officinalis* compared to other species is shown in purple, putative sex chromosome as identified recently⁸⁷ is shown in cyan.

B) Normalized coverage of sequencing data in *S. officinalis* chromosomes. **C)** Normalized coverage of short reads to female *S. esculenta* genome, reproduced from⁸⁷. Decrease in read coverage for chromosome 46 visible, the putative Z sex chromosome. Read depth was calculated from Illumina gDNA reads in windows of 500,000 bp and normalized to the median coverage of chromosome 1. Box plots showing median divergence (box dividing line), interquartile range (box), and 1.5 times the interquartile range (whiskers). Putative Z chromosome is highlighted in cyan.

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Reviewer #1 (Public review):

Summary:

This manuscript presents a high-quality, chromosome-level genome assembly of the European cuttlefish (*Sepia officinalis*), a representative species of the cephalopod lineage. Using state-of-the-art sequencing and scaffolding technologies -including PacBio HiFi long reads and Hi-C chromatin conformation capture - the authors deliver a genome assembly with exceptional contiguity and completeness, as evidenced by high BUSCO scores. This genome resource fills a significant gap in cephalopod genomics and offers a valuable foundation for studies in neurobiology, behavior, and evolutionary biology. However, there are several major aspects that need to be strengthened.

Major Revisions Recommended:

(1) Single-individual genome limitation

The genome assembly is based on a single individual, which appears to be male. While this approach is common in genome projects, it does not capture the full genetic diversity of the species. As *S. officinalis* exhibits a wide geographical range and possible population structure, future efforts (or discussion in this manuscript) should consider re-sequencing multiple individuals - of both sexes and from diverse geographic origins - to characterize population-level variation, sex-linked features, and structural polymorphisms.

(2) Limited experimental validation of chromosomal inferences

The study reports chromosome-scale scaffolding using Hi-C data and proposes a revised karyotype for *S. officinalis*. However, these inferences would be significantly strengthened by

orthogonal validation methods. In particular, fluorescence in situ hybridization (FISH) or karyotyping from cytogenetic preparations would provide direct confirmation of chromosome number and structural arrangements. The reliance solely on Hi-C contact maps for inferring chromosomal organization should be acknowledged as a limitation or supplemented with such validations.

(3) Shallow discussion of chromosomal evolution

The manuscript briefly mentions chromosomal number differences among cephalopods but does not explore their evolutionary or functional implications. A more thorough comparative analysis - linking chromosomal rearrangements (e.g., fusions, fissions) with ecological adaptation, life history, or neural complexity - would greatly enhance the impact of the findings. Referencing chromosomal dynamics in related taxa and possible links to behavioral innovations would contextualize these results more effectively.

(4) Underdeveloped gene family and pathway analysis

While the authors identify expansions in gene families such as protocadherins and C2H2 zinc finger transcription factors, the functional significance of these expansions remains speculative. The manuscript would benefit from:

- a) Functional enrichment analyses (e.g., GO, KEGG) targeting these gene families.
- b) Expression profiling across tissues or developmental stages to infer regulatory roles.
- c) Comparison with expression or expansion patterns in other cephalopods with known behavioral complexity (e.g., *Octopus bimaculoides*, *Euprymna scolopes*).
- d) Potential integration of transcriptomic or epigenomic data to support regulatory hypotheses.

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Reviewer #2 (Public review):

Summary:

This paper concerns an interesting organism, *Sepia officinalis*. However, in the opinion of this reviewer, the paper reads somewhat like a genome report. The authors have used 23x PacBio HiFi in conjunction with relatively low coverage (11x) Hi-C to scaffold the genome into a karyotype of 47 chromosomes. They have used a combination of short and long read RNA seq to annotate the genome in what looks like a very good annotation. The paper offers basic analyses of the Busco evaluation, some descriptive analyses of gene family and repeat content, and a bit more focused analysis on synteny among sequenced squids. Generally, the data will be useful.

Strengths:

This is a high-quality annotation, and the data ultimately will be useful to other researchers. I appreciate trying to understand what's happening between assemblies of *S. officinalis*.

Weaknesses:

I don't believe the data at hand makes a strong case for the argument of 47 chromosomes. This is my biggest sticking point with the paper, and it is for a few reasons:

- (1) The authors point to assembly differences between the DToL assembly and the one presented in the manuscript and seem to claim that DToL is incorrect. However, the DToL

assembly (xcSepOffi3.1) is based on much deeper HiFi and HiC coverage than the one at hand (51x and 80+x respectively). There are many things to try here, including:

- a) Downloading the DToL data and reassembling using a common pipeline.
- b) Downsampling the DToL data to similar coverage as what the authors have achieved.
- c) Combining your data and that of DToL for even deeper coverage (heterozygosity is low enough that I don't imagine this impeding things too badly).

(2) Looking at Figure 1, there appears to be a misjoin at chromosome 42. Looking carefully at Figure S1, that misjoin does not appear on any of the panels - this is confusing. Given the size of that chromosome and the authors' chromosome numbering, I'm guessing this is a manual merge (as it's larger than most of the chromosomes numerically close (40, 41, 43, etc). Further, staring closely at Figure 1, there appear to be cross-scaffold contacts between 42 and 43 and 42 and 44. Secondly there are contacts between 43 and 44. This bit of the assembly seems potentially problematic.

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Reviewer #3 (Public review):

Summary:

In this study, authors Simone Rencken and co-authors present and investigate the genome of the common cuttlefish *Sepia officinalis*.

Strengths:

The authors explain in a detailed yet concise manner the main steps for a genome assembly, with very robust methods for validation, and according to current best practices. In addition to the chromosomal assembly, the authors confirmed the presence of 47 chromosomes using Hi-C data and multiple species synteny. They also generated a comprehensive gene annotation, with assessments of gene completeness, providing a useful resource for the community of researchers interested in cuttlefish biology and comparative genomics.

Weaknesses:

While the study touches upon the subjects of gene content, TE activity, or species-level comparisons, the study does not provide in-depth investigations of these.

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